

The History and Overview of Java

Lecture 2

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Object Oriented Programing

- Paradigms: All computer programs consist of two elements
 - Code
 - Data

OOP Principles

- All object-oriented programming languages provide mechanisms that help you implement the object-oriented model. The mechanisms are:
 - Encapsulation
 - Inheritance
 - Polymorphism

Encapsulation

- Encapsulation is the mechanism that binds together code and the data it manipulates and keeps both safe from outside interference and misuse.

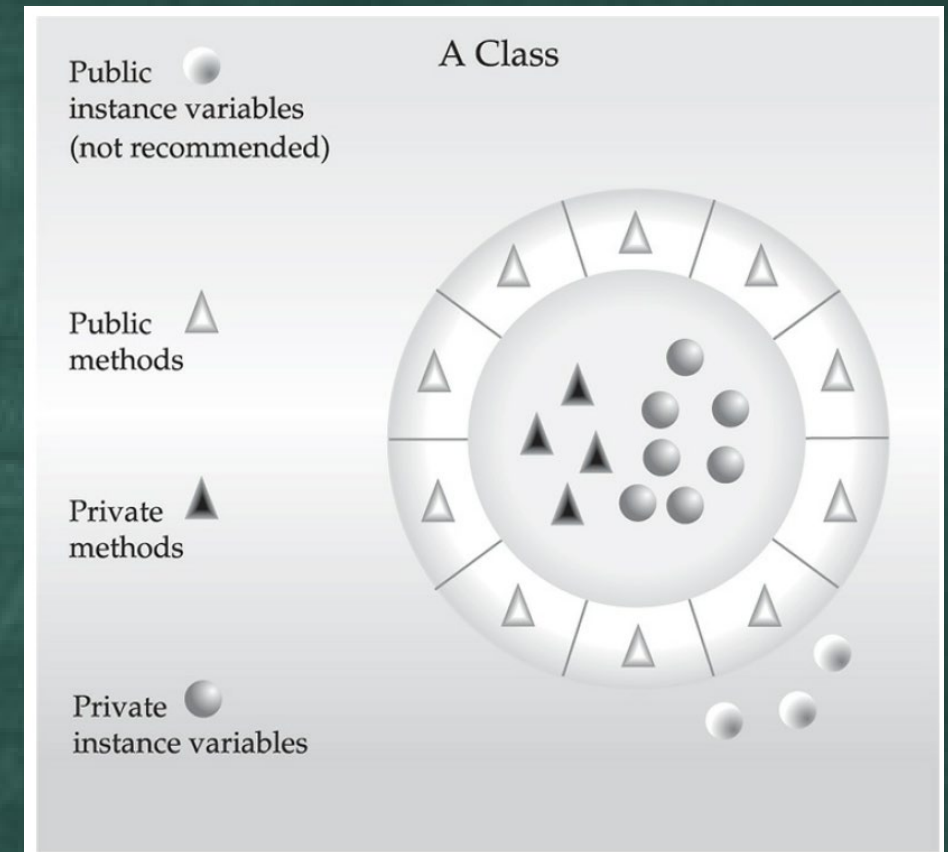


Fig. 2.1: Encapsulation: public methods can be used to protect private data.

Inheritance

- Inheritance is the process by which one object acquires the properties of another object. This is important because it supports the concept of hierarchical classification.

Inheritance in Java

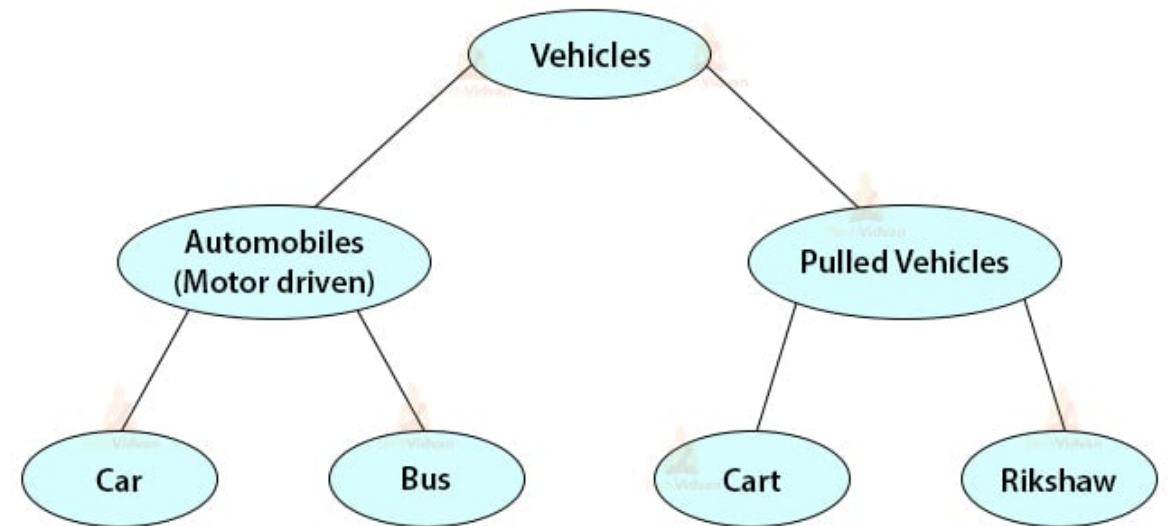


Fig. 2.2: Inheritance in Java

Polymorphism

- Polymorphism (from Greek, meaning “many forms”) is a feature that allows one interface to be used for a general class of actions

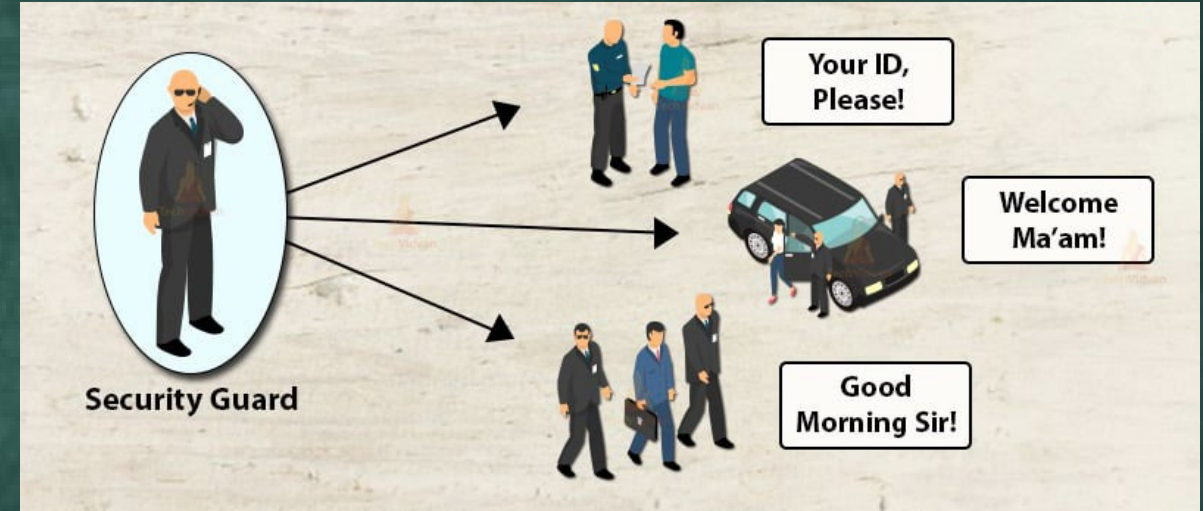


Fig. 2.3: Polymorphism in Java

What is Java?

- Java is a programming language and a platform. Java is a high level, robust, object-oriented and secure programming language.
- Java was developed by Sun Microsystems (which is now the subsidiary of Oracle) in the year 1995. James Gosling is known as the father of Java. Before Java, its name was Oak. Since Oak was already a registered company, so James Gosling and his team changed the name from Oak to Java.
- Platform: Any hardware or software environment in which a program runs, is known as a platform. Since Java has a runtime environment (JRE) and API, it is called a platform.

Your First Program in Java

```
1. // This is my first java program.  
2. / * This will print 'Hello, World!'  
3.   * as the output  
4.   */  
5. public class MyFirstJavaProgram {  
  
6.     public static void main(String []args) {  
7.         System.out.println("Hello, World!"); // prints Hello, World!  
8.     }  
9. }
```


Commenting Your Programs

- We insert comments to document programs and improve their readability. The Java compiler ignores comments, so they do not cause the computer to perform any action when the program is run.
- By convention, we begin every program with a comment indicating the figure number and filename. The comment in line 1

// This is my first java program.

Java also has traditional comments, which can be spread over several lines as in Line 2-4 using /* and */

/* This will print 'Hello, World!'

 * as the output

*/

Declaring a Class

```
public class MyFirstJavaProgram
```

- Every Java program consists of at least one class that you (the programmer) define. The class keyword introduces a class declaration and is immediately followed by the class name (**MyFirstJavaProgram**). Keywords (sometimes called reserved words) are reserved for use by Java and are always spelled with all lowercase letters.
- **Filename for a public Class:** A public class must be placed in a file that has a filename of the form **ClassName.java**, so class **MyFirstJavaProgram** is stored in the file **MyFirstJavaProgram.java**.

Class Names and Identifiers

- By convention, class names begin with a capital letter and capitalize the first letter of each word they include (e.g., `SampleClassName`).
- A class name is an identifier—a series of characters consisting of letters, digits, underscores (`_`) and dollar signs (`$`) that does not begin with a digit and does not contain spaces.
- Some valid identifiers are `Welcome1`, `$value`, `_value`, `m_inputField1` and `button7`.
- The name `7button` is not a valid identifier because it begins with a digit, and the name `input field` is not a valid identifier because it contains a space. Normally, an identifier that does not begin with a capital letter is not a class name.
- Java is case sensitive—uppercase and lowercase letters are distinct—so `value` and `Value` are different (but both valid) identifiers.

Class Body

- A left brace (as in line 5), {, begins the body of every class declaration. A corresponding right brace (at line 9), }, must end each class declaration. Lines 6-8 are indented.

```
1. // This is my first java program.
2. / * This will print 'Hello, World!'
3.   * as the output
4.   */
5. public class MyFirstJavaProgram {
6.     public static void main(String
       []args) {
7.         System.out.println("Hello,
       World!"); // prints Hello, World!
8.     }
9. }
```

Declaring a Method

- Line 6 is an is an end-of-line comment indicating the purpose of lines 7-8 of the program. Line 7 is the starting point of every Java application.
- The parentheses after the identifier main indicate that it's a program building block called a method.

```
1. // This is my first java program.  
2. /* This will print 'Hello, World!'  
3.    * as the output  
4. */  
5. public class MyFirstJavaProgram {  
6.     public static void main(String []args) {  
7.         System.out.println("Hello,  World!");  
           // prints Hello, World!  
8.     }  
9. }
```


Declaring a Method cont.

- Java class declarations normally contain one or more methods. For a Java application, one of the methods must be called main and must be defined as shown in line 7; otherwise, the Java Virtual Machine (JVM) will not execute the application.
- Methods perform tasks and can return information when they complete their tasks. For now, simply mimic main's first line in your Java applications. In line 7, the String[] args in parentheses is a required part of the method main's declaration—we discuss this later

```
1. // This is my first java program.  
2. /* This will print 'Hello, World!'  
3.    * as the output  
4.    */  
5. public class MyFirstJavaProgram {  
6.     public static void main(String []args) {  
7.         System.out.println("Hello, World!"); // prints Hello, World!  
8.     }  
9. }
```


Declaring a Method cont.

- The left brace in line 7 begins the body of the method declaration. A corresponding right brace must end it (line 8). Line 7 in the method body is indented between the braces.

```
1. // This is my first java program.  
2. /* This will print 'Hello, World!'  
3.    * as the output  
4. */  
5. public class MyFirstJavaProgram {  
6.     public static void main(String  
7.         []args) {  
8.         System.out.println("Hello,  
9.         World!"); // prints Hello, World!  
10.    }  
11. }
```

Performing Output with System.out.println();

- Line 7 instructs the computer to perform an action—namely, to display the characters contained between the double quotation marks.
- The System.out object—which is predefined for you—is known as the standard output object.
- Method System.out.println displays (or prints) a line of text in the command window.
- The entire line 7, including System.out.println, the argument "Hello, World!" in the parentheses and the semicolon (;), is called a statement.

```
1. // This is my first java program.
2. /* This will print 'Hello, World!'
3.    * as the output
4. */
5. public class MyFirstJavaProgram {
6.     public static void main(String []args) {
7.         System.out.println("Hello, World!"); // prints
           Hello, World!
8.     }
9. }
```

C++ vs Java

TOPIC	C++	Java
Memory Management	Use of pointers, structures, union	No use of pointers. Supports references, thread and interfaces.
Libraries	Comparatively available with low-level functionalities	Wide range of classes for various high-level services
Multiple Inheritance	Provide both single and multiple inheritance.	Multiple inheritances is partially done through interfaces
Operator Overloading	Supports operator overloading	It doesn't support this feature
Program Handling	Functions and variables can reside outside classes.	Functions and variables reside only in classes, packages are used.
Portability	Platform dependent, must be recompiled for different platform	Platform independent, byte code generated works on every OS.
Thread Support	No built-in support for threads, depends on libraries.	It has built-in thread support.

Thank You